TEMPO: Regime-Aware Operator Learning with Locally Adapted POD Bases

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2026

Problem Statement

Task

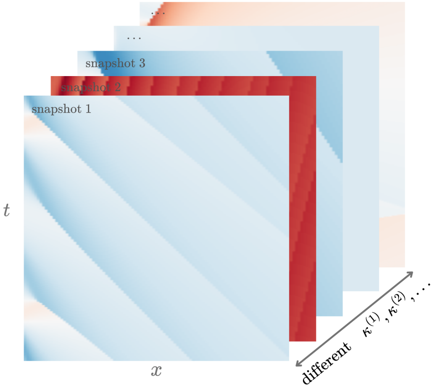
Given $\mathcal{D} = \{(u_0^{(i)}, \kappa^{(i)}, s^{(i)})\}_{i=1}^{N}$, construct $\hat{\mathcal{G}} \approx \mathcal{G}\colon (u_0, \kappa) \mapsto s$ accurate across all qualitative regimes, without solving the PDE.

Problem

Existing methods learn a single basis for all κ. When solutions change character fundamentally across parameter values — shock structures at $\nu{=}0.001$ versus smooth diffusion at $\nu{=}1.0$ — the basis must compromise between incompatible geometries, concentrating error precisely where accuracy matters most.

Proposal: discover latent regimes unsupervised and build a dedicated basis per regime —**TEMPO**

TEMPO: Central Idea — Regime-Aware Operator Learning



Offline: EM-GMM $\Rightarrow$ M bases $\{\Phi_m\}$

Online: $\hat{s} = \sum_m w_m \left[\phi_0^{(m)} + \sum_k b_{mk}\, \phi_k^{(m)} \right]$

$$\mathcal{G} : \mathcal{U} \times \mathcal{K} \to L^2(\Omega \times \mathcal{T}), \quad (u_0, \kappa) \mapsto s$$

Learn $\hat{\mathcal{G}} \approx \mathcal{G}$ from $\mathcal{D}$ without solving the PDE.

TEMPO

Discover M latent regimes via EM-GMM. Build a dedicated basis per regime *(POD or NeuralPOD)*. Route new inputs at inference with a learned gating network.

Related Work

DeepONet Lu et al., 2019

Branch encodes input function; trunk encodes query location. Universal approximation of nonlinear operators.

POD-DeepONet Lu et al., 2022

Fixed POD basis replaces the trunk network. Compact, interpretable, data-efficient.

Neural-POD Mou et al., 2026

Nonlinear orthogonal basis via greedy residual minimization. Captures structure beyond any linear subspace.

Local Reduced-Order Bases Amsallem et al., 2012

Snapshot space partitioned into clusters; separate POD basis per cluster. Hard assignment, offline only.

EM Algorithm Dempster, Laird, Rubin, 1977

Maximum likelihood with latent variables. Monotone log-likelihood convergence at every iteration.

TEMPO combines Neural-POD bases with soft EM regime discovery, integrating local basis construction and operator learning end-to-end.

Problem Statement

Hypothesis. Each trajectory arises from one of M latent regimes; labels z_i are unobserved.

$$z_i \sim \mathrm{Categorical}(\pi_1, \dots, \pi_M)$$

$$\left(s^{(i)} \mid z_i{=}m\right) \sim \mathcal{N}\!\left(\hat{s}^{(i)}_m,\ \sigma^2 W^{-1}\right)$$

$$\hat{s}^{(i)}_m(y) = \phi^{(m)}_0(y) + \sum_{p=1}^{P_m} \lambda^{(m,i)}_p \, \phi^{(m)}_p(y)$$

z_i	regime label (latent)
π_m	prior weight
σ^2	assignment softness
$\phi^{(m)}_p$	spatiotemporal mode
$\lambda^{(m,i)}_p$	scalar amplitude

Find $\Phi^*, \pi^*, \sigma^{2*}$ maximising the observed-data log-likelihood (EM).

$$\varepsilon = \frac{1}{N_{\text{test}}} \sum_i \frac{\left\| s^{(i)} - \hat{s}^{(i)} \right\|}{\left\| s^{(i)} \right\|}$$

Solution

POD (linear SVD) $\rightarrow$ POD-DeepONet | **TEMPO(POD)**
NeuralPOD (greedy MLP) $\rightarrow$ NeuralPOD-DeepONet | **TEMPO(NeuralPOD)**

Phase 1 — Offline: obtain $\boldsymbol{\lambda}^{(i)}$

POD: $S \approx U_K \Sigma_K V_K^\top, \quad \lambda_k^{(i)} = \langle s^{(i)}, \phi_k \rangle$

NeuralPOD:

$$\min_{\phi_k,\, \{\lambda_k^{(i)}\}} \frac{1}{N} \sum_{i=1}^{N} \left\| r^{(k-1,i)} - \lambda_k^{(i)} \phi_k \right\|^2$$

TEMPO: same, but γ_{im}-weighted. For POD:

$$\bar{s}_m = \textstyle\sum_i \gamma_{im} s^{(i)}, \qquad S_w = \mathrm{diag}(\sqrt{\gamma_{\cdot m}})\, \tilde{S}$$

Phase 2 — Online: train BranchNet on $\boldsymbol{\lambda}^{(i)}$

POD-DeepONet / NeuralPOD-DeepONet:

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \left\| \mathcal{B}_\theta(u_0^{(i)}, \kappa^{(i)}) - \boldsymbol{\lambda}^{(i)} \right\|^2$$

TEMPO (GatingNet + M BranchNets):

$$\mathcal{L}_{\text{online}} = \mathcal{L}_{\text{data}} + \lambda_{\text{KL}}\, \mathcal{L}_{\text{KL}} - \lambda_H\, \mathcal{L}_{\text{ent}}$$

The offline $\boldsymbol{\lambda}^{(i)}$ are the regression targets for the online BranchNet in all four methods.

Computational Experiment

Dataset: 1D Burgers' equation (PDEBench)

$$u_t + u\,u_x = \nu\,u_{xx}, \quad x \in (-1, 1), \quad t \in (0, 2]$$

$\nu \in \{0.001, 0.01, 0.1, 1.0\}$ — three orders of magnitude. $9{,}500$ trajectories per ν $\Rightarrow$ $38{,}000$ total.
Grid: $N_x{=}1{,}024$, $N_t{=}101$.

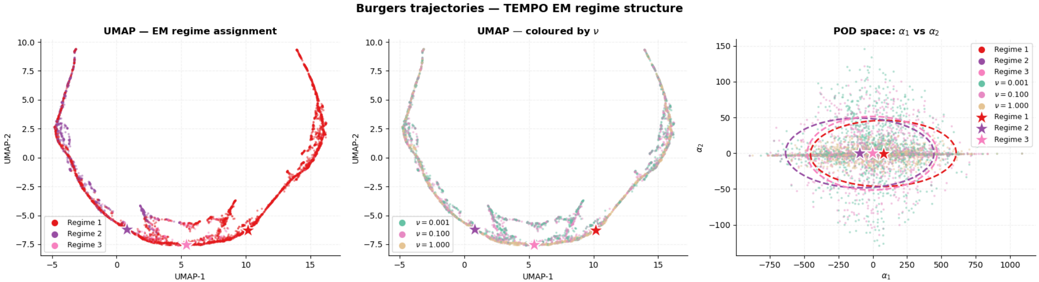
Three verification claims:

1) Single-regime bases concentrate error at regime extremes
2) EM regimes cross ν reflects κ and geometry
3) TEMPO outperforms all baselines across the full ν range

Methods compared:

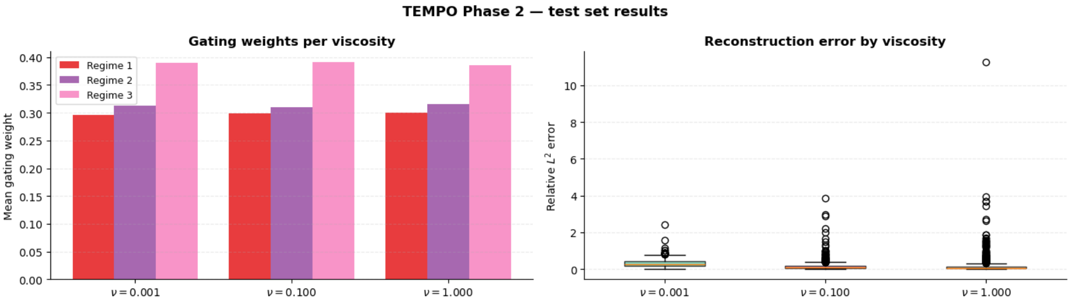
DeepONet	baseline
POD-DeepONet	linear, 1 regime
NeuralPOD-DeepONet	nonlinear, 1 regime
TEMPO(POD)	linear, M regimes
TEMPO(NeuralPOD)	nonlinear, M regimes

Error Analysis



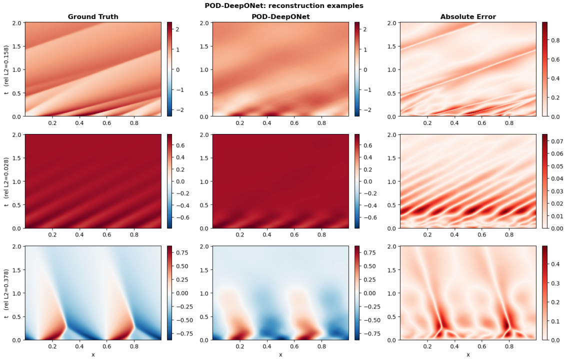
M chosen by minimising BIC and mean assignment entropy $H = -\frac{1}{N}\sum_{i,m} \gamma_{im} \log \gamma_{im}$: optimal at $M{=}3$, one basis per qualitative regime. Regimes *cross ν isolines* — EM partitions by solution geometry, not by κ.

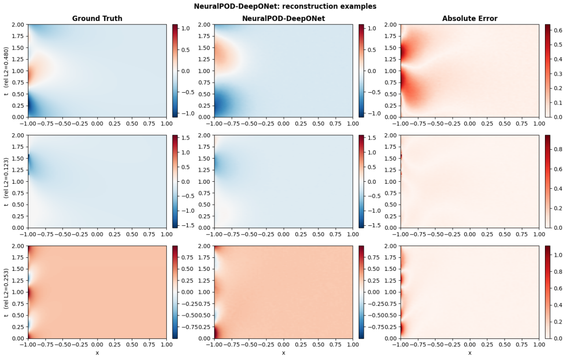
Results: gating and error



Left: mean gating weight $w_m(u_0, \kappa)$ per viscosity — each regime dominates a distinct ν band. *Right:* relative L^2 error distribution per viscosity on the test set.

Results: reconstruction





POD-DeepONet: ground truth / prediction / error

NPOD-DeepONet: ground truth / prediction / error

TEMPO achieves mean relative L^2 error ≈ 0.20 versus 0.2130 (POD-DeepONet) and 0.2444 (NeuralPOD-DeepONet) — and unlike the baselines, maintains this accuracy *simultaneously across all viscosity regimes*.

Conclusion

Contributions

- TEMPO: offline EM-GMM regime discovery + dedicated basis per regime + gated online operator
- NeuralPOD-DeepONet as a strong nonlinear single-regime baseline
- EM regimes reflect solution geometry and κ labels

Main result

Regime-specific bases achieve lower reconstruction error at similar mode count; the gated operator outperforms all single-basis baselines

Future work

- Parameter-conditioned modes $\phi_k^{(m)}(\cdot;\kappa)$
- Extension to 2D (Darcy flow)
- Convergence theory for the GEM update